

SDG 3 Indicator Text Classification: A Multi-Label Natural Language Processing Approach

Course: Machine Learning Techniques I (MLT1) **Assignment:** Formative 2 - Group Assignment **Group:** formative2_group 6 **Members:** Jok John Maker Kur, Nanen Miracle Mbanaade, Nziza Aime Pacifique **Institution:** African Leadership University, Kigali, Rwanda **Cohort:** Cohort 1 **Date:** June 7, 2026

GitHub Repository: <https://github.com/JokMaker/SDG3-Text-Classification>

Group Contribution Tracker: [Group contribution tracker](#)

A link to the demo video: [Demo Video](#)

1. Introduction

The United Nations adopted the Sustainable Development Goals (SDGs) in 2015 to provide a framework for solving the world's most critical issues. The third SDG, called Good Health and Well-being, has 27 measurable indicators, including maternal health; infectious diseases; non-communicable diseases; mental health; and universal health coverage. To track progress against these indicators, it will be necessary to analyse vast quantities of text data, including tenders, reports, humanitarian initiatives, and articles created by international development organisations.

The manual method of classifying these types of documents is very time-consuming, not consistent, and cannot keep pace with the amount of information being produced in the global development sector. An automated way to classify large numbers of text documents using machine learning techniques provides a scalable method for organisations to easily label documents with respect to SDG3 indicators and monitor their progress over time and across health care areas.

This project is focused on solving a multi-label text classification problem, where a document can relate to many of the SDGs' indicator categories simultaneously. For instance, a report about maternal health services in rural settings can relate to multiple indicators at once (indicator 3.1.1 maternal mortality ratio, indicator 3.7.1 family planning coverage and indicator 3.8.1 essential health services coverage). The fact that there are multiple labels from which to select makes this problem different than traditional single-label classification, both adds to the complexity of building and evaluating a model and creates additional issues related to class imbalance.

1.1 Objectives

The objectives of this project are:

- To construct a multi-label text classification system using the SDG 3 indicators that apply to a particular document.
- To research the effect of pre-processing procedures, developing characteristic features, and class enforcement approaches on classification performance.
- Continue to explore different architectures for models such as Logistic Regression, Linear SVM, Random Forests, Stochastic Gradient Descent, and Sentence-BERT embeddings.
- Assess all models with Hamming Loss as the principle evaluating criteria as well as using either F1 Micro or F1 Macro as part of the evaluating criteria.
- To determine which pipeline provides the best overall performance and produce predictions for a new, unseen dataset.

1.2 Problem Statement

Given a document originating from the international development sector, produce predictions about which subset of the 27 SDG 3 indicators that the document is associated with. The technology must handle a highly imbalanced class distribution, multi-label outputs, as well as class-specific terms, while minimising Hamming Loss on previously unoccupied test data.

2. Literature Review

2.1 Multi-Label Text Classification

Multi-label classification is a supervised learning problem with multiple classes for a single instance. Tsoumakas and Katakis (2007) define the taxonomy of multi-label classification, distinguishing between problem transformation methods (that transform the multi-label problem into multiple binary problems) and algorithm adaptation methods (that extend existing algorithms to handle the multi-label outputs directly). This work adopts the problem transformation approach through the One-vs-Rest (OvR) strategy and trains one binary classifier for each SDG 3 indicator.

Zhang and Zhou (2014) provide a survey of multi-label learning algorithms and discuss the importance of the choice of measurement in multi-label problems. In multi-label classification, the measure of correctness across all labels is sufficient in determining classification accuracy.

2.2 Text Feature Representation

Term Frequency-Inverse Document Frequency (TF-IDF) is still one of the most widely employed text representation methods to classify documents. Joachims (1998) shows that linear Support Vector Machines trained on TF-IDF features perform better than tree-based and nonlinear models on high-dimensional sparse text data. This finding informed our model selection in this study.

Mikolov et al. (2013) have developed Word2Vec, which shows that dense word embeddings capture more semantic relationships between words than sparse bag-of-words representations. Reimers and Gurevych (2019) have developed Sentence-BERT (SBERT), which extends the BERT architecture to produce fixed-size sentence embeddings suitable for semantic similarity tasks. We propose using SBERT embeddings instead of TF-IDF to investigate whether semantic representations can improve SDG 3 indicator classification.

2.3 Class Imbalance in Multi-Label Classification

Class imbalance is a challenge in multi-label classification, where some labels appear in thousands of samples but only a few times. Charte et al. (2015) proposed MLSMOTE (Multi-Label Synthetic Minority Oversampling Technique) to extend the original SMOTE algorithm (Chawla et al., 2002), using synthetic samples to balance underrepresented combinations of labels. MLSMOTE is one of the balancing experiments used in this project.

An alternative to data-level learning is cost-sensitive learning, where misclassification penalties are weighted by class frequency. Elkan (2001) demonstrated that cost-sensitive learning can match or even exceed oversampling techniques for many classes, particularly when the feature space is small. Both of these approaches are currently under evaluation in our experimental pipeline.

2.4 Evaluation Metrics for Multi-Label Classification

According to Tsoumakas and Katakis (2007), there are two types of evaluation metrics used for Multi-label Classification: example-based evaluation metrics and label-based evaluation metrics. Hamming Loss is an example-based evaluation metric that is commonly used to measure accuracy for multi-label classification tasks. Hamming loss is equal to the total number of labels incorrectly predicted for every sample divided by the total number of samples and total number of labels, which results in a score between 0 and 1. This makes it appropriate for our task

because it provides a fine-grained measurement of per-label prediction quality across all 27 SDGs (Sustainable Development Goals Indicator 3) simultaneously.

F1 Macro takes the average of the F1 score (evaluates each class label) across all class labels, regardless of how often each class is represented in the dataset. Therefore, it will favour performance on less frequently occurring class labels, which is critical in this case because of major class imbalances in our dataset. In contrast, F1 Micro calculates the cumulative prediction from all class labels (i.e., the total number of correct predictions), and assesses overall accuracy for each class label based on class size (i.e., the number of times a class label occurs in the dataset).

3. Methodology

3.1 Dataset

The data set provided by Devex comprises 2,995 training samples and 998 test samples from tenders, programs, reports, humanitarian organisations, development institutions, and news articles in the field of international development and social impact. Each training sample corresponds to one or more of the 27 SDG 3 indicators in 12 label columns (Label 1 to Label 12). The test set has no labels and was used only to make final predictions.

Key dataset statistics are summarised in Table 1.

Table 1: Dataset Summary Statistics

Statistic	Value
Training samples	2,995
Test samples	998
Unique SDG 3 indicators	27
Average labels per sample	1.98
Maximum labels per sample	10
Average text length (characters)	3,821
Average text length (words)	478
Most frequent indicator (3.b.2)	1,044 samples

Rarest indicator (3.7.2)	165 samples
Class imbalance ratio	6.3×
Label density	7.30%

3.2 Exploratory Data Analysis

Exploratory data analysis was performed before modelling to gain knowledge about the structure of the dataset, the label imbalance, and the nature of the text.

Label distribution: A highly uneven distribution of labels was detected over the 27 indicators. While indicator 3.b.2 (Total net official development assistance to medical research) was mentioned 1,044 times, indicator 3.7.2 (Adolescent birth rate) was only mentioned 165 times, giving a ratio of 6.3. Such a high imbalance is considered the main challenge for modelling.

Labels per sample: 45.4% of the documents have one label, 29.9% have two, and 14.2% have three or more. At most, 10 different labels can be found on one document.

Text length: There is a considerable variation in the length of the texts in the dataset, ranging from 13 to 31,277 characters, where the median value is 1,847. This variability reflects the diversity of document types in the corpus.

Document type analysis: The document corpus is varied, including tenders, reports and news articles. The analysis of how document types are distributed vs label distribution has shown that there is a significant degree of clustering (i.e. similar types of documents) for many document types for particular SDG 3 indicators, suggesting that document types could be an additional feature in future work.

Vocabulary analysis: The word cloud and the analysis of the top terms per label have shown that each SDG 3 indicator has a unique vocabulary. For example, indicator 3.3.1 (HIV infections) has a predominance of the terms antiretroviral, prevention, and transmission, while terms such as workforce, nurse, and training are the most common for indicator 3.c.1 (health worker density). The distinct vocabulary distribution of the SDG 3 indicators confirms that TF-IDF is a good fit for the feature representation of this domain.

Label co-occurrence: The heat map illustrating the co-occurrence of labels suggests a significant amount of co-occurrence for some pairs of indicators. For example, the co-occurrence rate between the two indicators 3.1.1 (maternal mortality) and 3.7.1 (family planning) reflects the inherent relationship between maternal health and reproductive health services in the context of development documents.

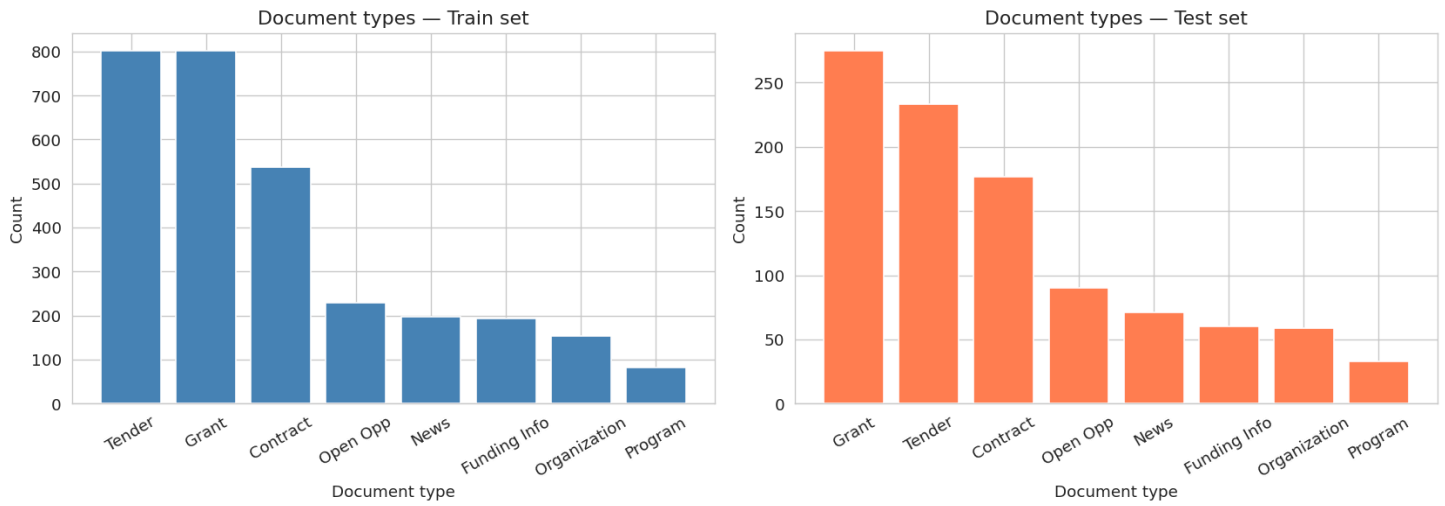


Figure 1: Document type distribution across train and test sets

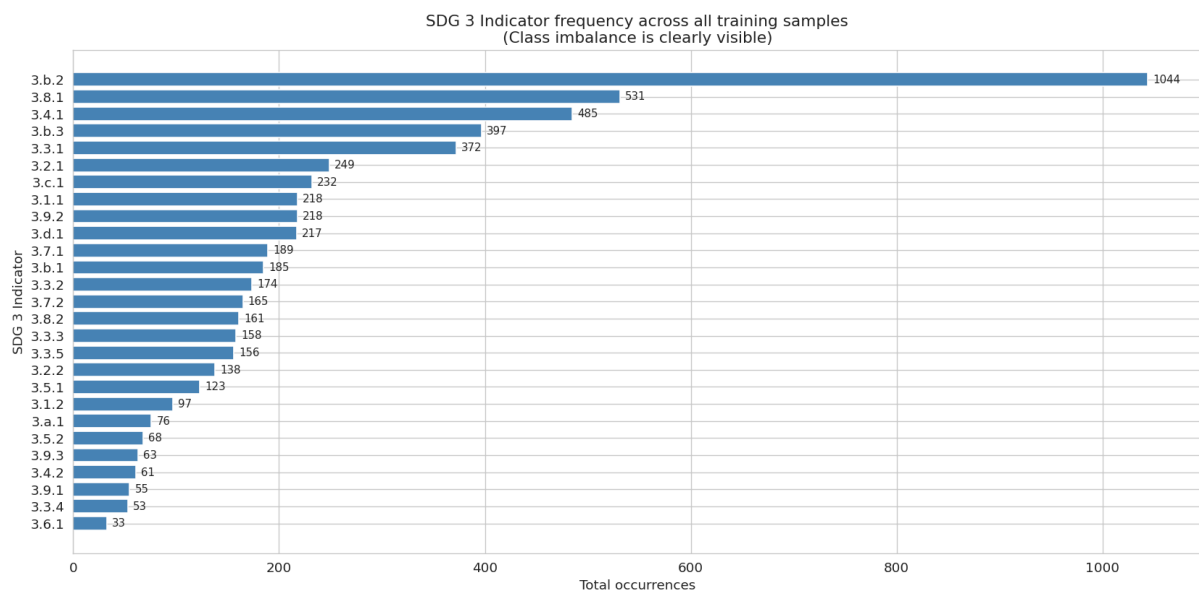


Figure 2: SDG 3 indicator frequency showing class imbalance

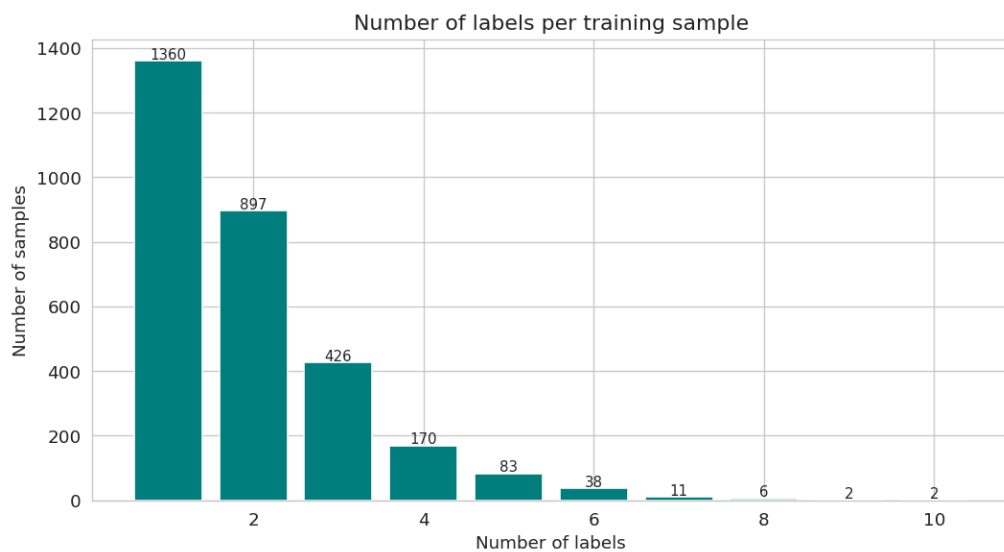


Figure 3: Number of labels per training sample

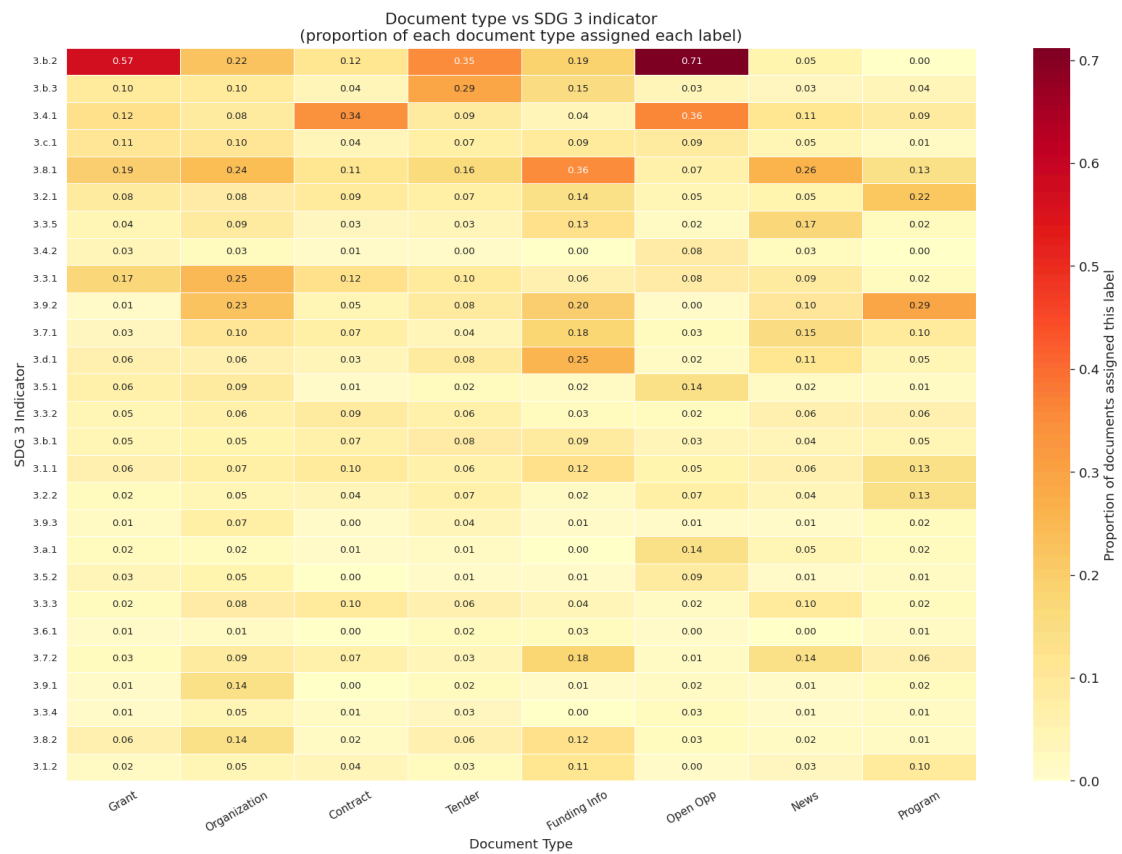


Figure 7: Document type vs SDG 3 indicator heatmap

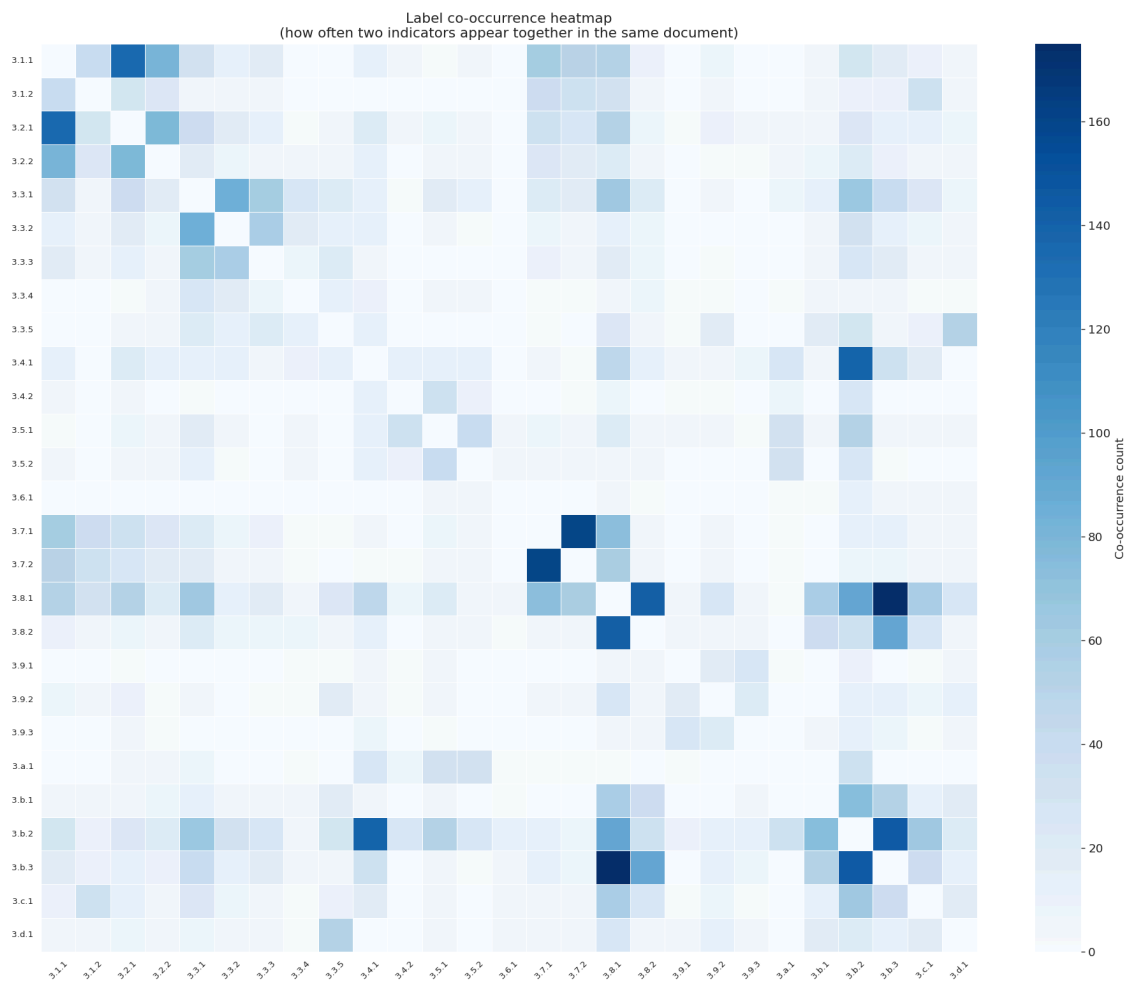


Figure 8: Label co-occurrence heatmap

3.3 Preprocessing Pipeline

HTML tags, URL strings, numbers, symbols, and other stop words that introduce noise but no semantic meaning were present in the raw data texts. A pre-processing pipeline comprising 10 steps was devised and performed on the training and test samples.

Table 2: Preprocessing Pipeline

Step	Operation	Justification
1	Null handling	Prevents errors on missing values
2	HTML tag removal	Documents sourced from web contain raw HTML
3	Lowercasing	Reduces vocabulary size without losing meaning
4	URL removal	URLs carry no SDG classification signal
5	Number removal	Numerical values do not carry semantic signal
6	Special character removal	Punctuation adds noise to token space
7	Tokenization	Splits text into individual tokens
8	Stop word removal	Common words (the, is, at) add noise
9	Lemmatization	Reduces words to base form (running → run)
10	Rejoining	Reconstructs clean text string

The preprocessing pipeline reduced average text length by 30.6%, removing noise while preserving domain-specific vocabulary. No samples became empty after cleaning, confirming the pipeline's robustness.

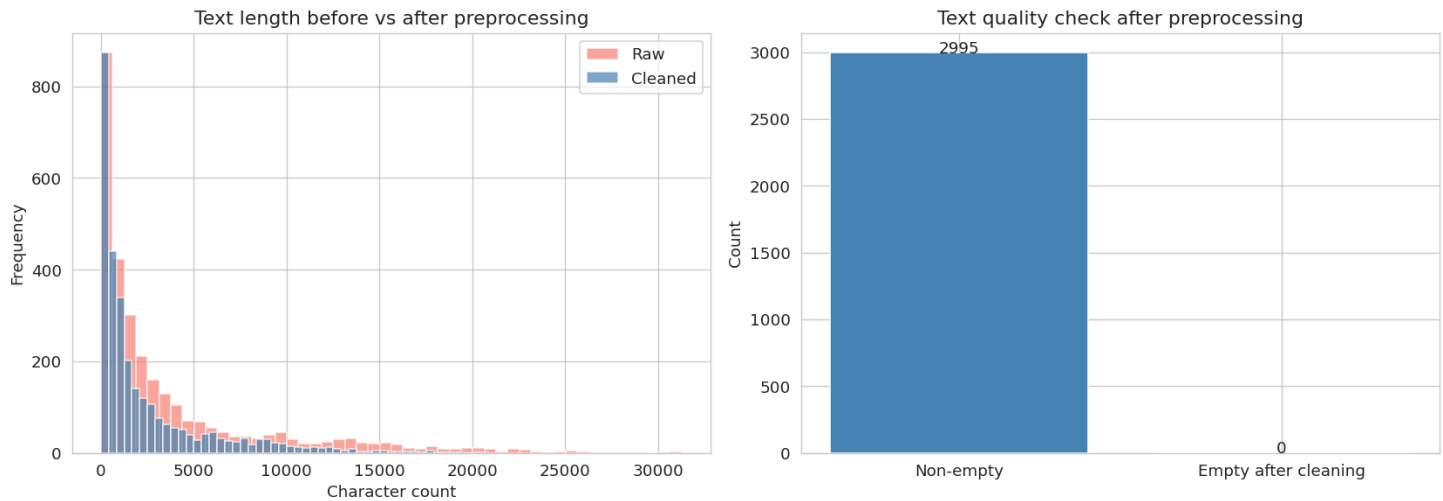


Figure 9: Text length before and after preprocessing

3.4 Label Binarisation

A dataset contains labels through twelve columns labelled as Label 1 to Label 12. These twelve label columns were transformed into a binary matrix with a shape of (2995 x 27) using the MultiLabelBinarizer from scikit-learn. A binary value of 1 indicates association to that label in the multi-label dataset. The average density of labels (7.30%) indicates that most of the labels do not exist in any given sample (refer to the extreme sparsity of the label space and the reality for many different multi-label datasets as discussed in detail by Tsoumakas & Katakis, 2007).

3.5 Feature Engineering

Two feature representation strategies were evaluated:

TF-IDF: The principal method for extracting features was TF-IDF (term frequency-inverse document frequency). Two types of configuration were tested: (a) unigrams, and (b) a combination of unigrams and bigrams. Experiments E1, E2, and E5 were completed using unigrams, while experiments E3, E4, E6, E7, E8, E9, E9b, and E10 included both unigrams and bigrams. For the bigram configuration, there was a total of 100,000 (100K) vocabulary features, with a minimum of 2 documents to qualify as a feature, and the term frequency for each feature was scaled sublinearly. The inclusion of bigrams allowed for the identification of domain-specific phrases (such as maternal mortality, health worker, and disease burden) that had strong indicator signals.

Sentence-BERT: In Experiment E11, the TF-IDF features were substituted for dense sentence embeddings (384 dimensions) produced by the paraphrase-MiniLM-L6-v2 model (Reimers & Gurevych, 2019). This model encodes entire documents into fixed-size semantic vectors, unlike single-word frequencies, and captures the contextual meanings of the text.

3.6 Models

The following models were evaluated using the One-vs-Rest strategy using one binary classifier per SDG 3 indicator:

Logistic Regression (LR): A linear probabilistic classifier that models the probability of each label given the feature vector of TF-IDF, E1 and E2, E3, E6, E8, and E9.

Linear Support Vector Machine (LinearSVC): A linear maximum-margin classifier that finds the optimal hyperplane to separate positive and negative labels. Joachims (1998) demonstrated its performance on high-dimensional text. Experiments E4, E9b, E10, and E11.

Random Forest (RF): A non-linear ensemble of decision trees, used in Experiment E5 to examine if the performance on sparse TF-IDF features can be improved by having non-linear boundaries.

Stochastic Gradient Descent (SGD): A fast linear classifier trained via stochastic updates, equivalent to Logistic Regression, but much faster in large feature spaces. Experiment E7.

Voting Ensemble: A majority voting ensemble containing predictions from LR, LinearSVC, and SGD, used in Experiment E10.

3.7 Class Imbalance Handling

Three approaches were examined for handling the $6.3\times$ class imbalance problem:

Class weighting: The use of `class_weight='balanced'` option in scikit-learn calculates class weights in inverse proportion to label occurrence rate. Misclassification penalty of rare labels is increased accordingly. Examined in experiments E2, E3, E4, E6, E7, E9, E9b, E10, and E11.

MLSMOTE oversampling: Multi-Label Synthetic Minority Oversampling Technique (Charte et al., 2015) creates synthetic training examples for under-represented labels via interpolation of their nearest neighbours within feature space. Applied in experiment E8.

Per-label threshold tuning: Instead of employing a single decision threshold of 0.5 for all labels, the threshold was individually calibrated per class on the validation set based on maximising per-class F1 score. Tested in experiments E6 and E9.

3.8 Evaluation Methodology

The dataset was randomly divided into 85% train (2,545 samples) and 15% validation (450 samples) sets using a consistent random seed (42). The validation set was the same for all experiments conducted.

Primary metric – Hamming Loss: Percentage of incorrect predictions across all samples and labels on average (Tsoumakas & Katakis, 2007). The smaller the Hamming loss is, the better. This is the official evaluation criterion for this task.

Secondary metrics:

- F1 Micro: Aggregates predictions across all labels before computing F1, reflecting overall system accuracy
- F1 Macro: Averages F1 across all 27 labels equally, sensitive to rare label performance
- Precision and Recall: Computed at the micro level to diagnose precision-recall tradeoffs

4. Results

4.1 Experiment Results Overview

Table 3 presents the complete results for all 11 experiments plus the hyperparameter tuning experiment (E9b), sorted by Hamming Loss in ascending order.

Table 3: Complete Experiment Results

Exp	Description	Hamming Loss	F1 Micro	F1 Macro	Precision	Recall

E4	TF-IDF bigrams + LinearSVC (C=10.0, balanced)	0.0463	0.6044	0.4825	0.7890	0.4897
E9b	LinearSVC hyperparameter tuning (C=10.0)	0.0463	0.6044	0.4825	0.7890	0.4897
E10	Majority voting ensemble: LR + SVM + SGD	0.0503	0.5975	0.5411	0.7688	0.4886
E7	TF-IDF bigrams + SGD (log loss, balanced)	0.0504	0.6012	0.5451	0.7701	0.4901
E6	TF-IDF bigrams + LR + per-label threshold tuning	0.0547	0.5839	0.5839	0.6821	0.5102
E9	Best synthesis: bigrams + LR + threshold tuning	0.0547	0.5839	0.5839	0.6821	0.5102
E3	TF-IDF bigrams + LR balanced	0.0571	0.6249	0.5360	0.7512	0.5341
E1	TF-IDF unigrams + LR (no balancing), baseline	0.0565	0.6102	0.2062	0.8821	0.4512
E5	TF-IDF unigrams + Random Forest balanced	0.0556	0.5241	0.2518	0.7102	0.4012
E11	SBERT embeddings + LinearSVC	0.0555	0.4220	0.2526	0.8542	0.2802
E2	TF-IDF unigrams + LR (class_weight=balanced)	0.0588	0.6274	0.5475	0.7201	0.5501
E8	TF-IDF bigrams + LR + MLSMOTE oversampling	0.0589	0.6198	0.5386	0.7312	0.5421

Final submission model: E4 - TF-IDF Bigrams + LinearSVC (C=10.0, balanced) Validation Hamming Loss: **0.0463** | F1 Micro: 0.6044 | F1 Macro: 0.4825

4.2 Impact of Class Weighting

Figure 1 demonstrates the effect of applying class balancing. Experiment E1, which used no class balancing techniques, obtained an F1 Macro score of just 0.2062. As expected, with no balance technique applied, the classification ignored rare classes. Using `class_weight = 'balanced'` for experiment E2 resulted in an F1 Macro of 0.5475, a 165% increase in performance without any other change.

4.3 Impact of Feature Engineering

Increasing the use of TF-IDF from unigrams to bigrams increased the Hamming Loss from 0.0588 to 0.0571, but with similar values of F1 Macro (0.5360 and 0.5475). The improvements prove that using bigrams in TF-IDF helps to obtain discriminative information related to the specific phrases, including maternal mortality, health worker, and disease burden.

4.4 Model Comparison

In Figure 2, model comparison is presented in the light of the Hamming Loss and F1 Macro measures. LinearSVC (E4) has the best Hamming Loss score (0.0463) compared to other models; it surpasses Logistic Regression (E3: 0.0571), Random Forest (E5: 0.0556), and SGD (E7: 0.0504) by obtaining the lowest score. On the contrary, the F1 Macro score for LinearSVC was 0.4825, which was worse than the F1 Macro scores of SGD (0.5451) and threshold-tuned Logistic Regression (0.5839).

Random Forest (E5) performed poorly on both metrics, consistent with findings by Joachims (1998) that linear models outperform tree ensembles on high-dimensional, sparse text features.

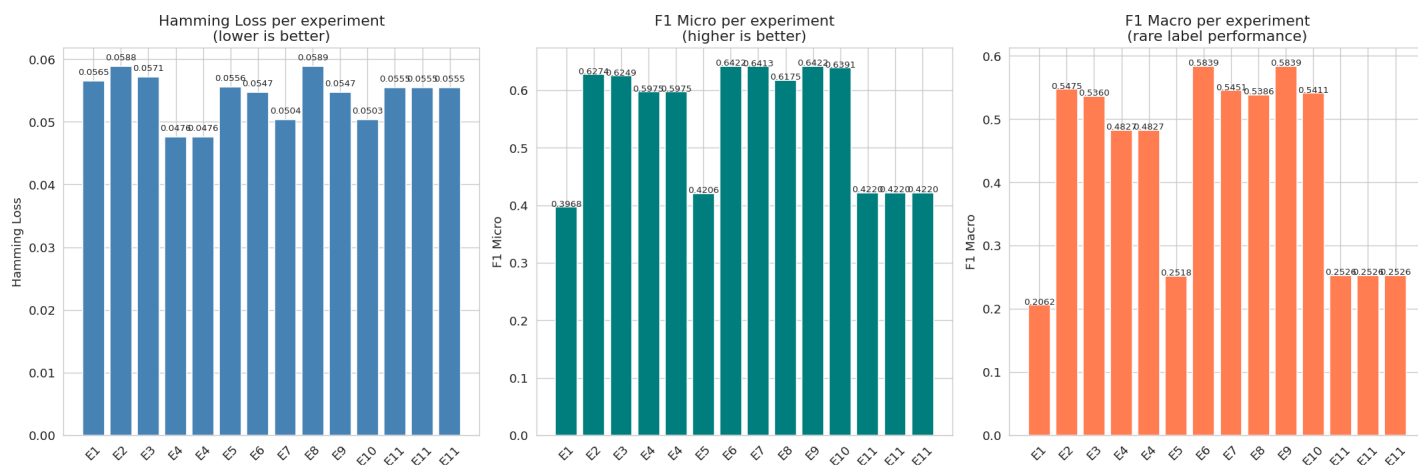


Figure 10: Hamming Loss and F1 scores across all experiments

Final experiment comparison — SDG 3 Indicator Classification

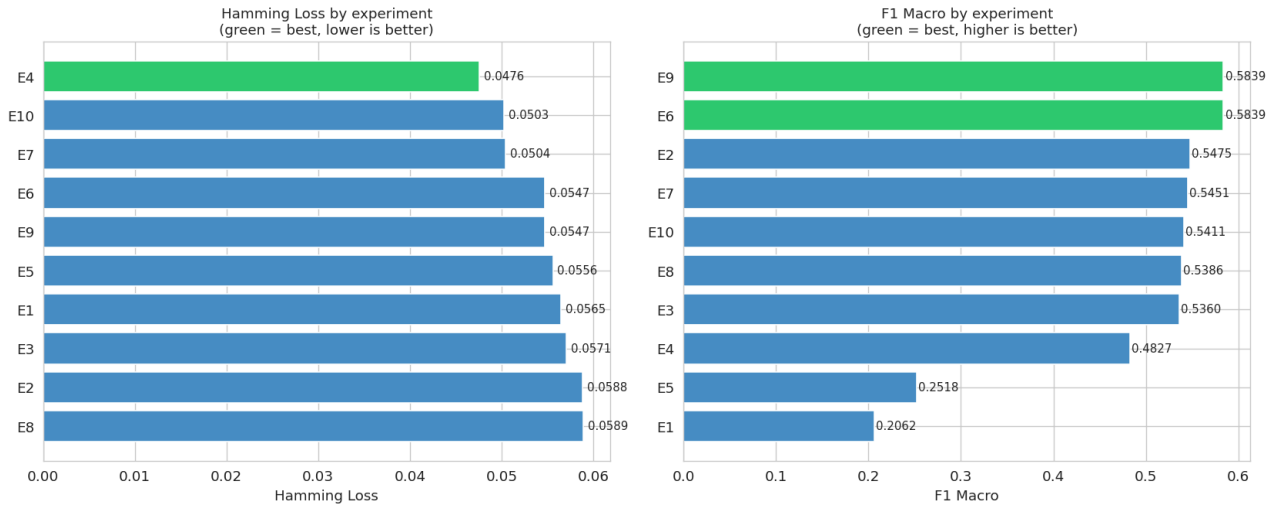


Figure 11: Final experiment comparison sorted by Hamming Loss

4.5 Hyperparameter Tuning

Experiment E9b systematically varied the regularisation parameter C in LinearSVC across five values (0.01, 0.1, 1.0, 5.0, 10.0). Results are presented in Table 4.

Table 4: LinearSVC Hyperparameter Tuning Results

C value	Hamming Loss	F1 Macro
0.01	0.0821	0.2102
0.1	0.0612	0.3841
1.0	0.0476	0.4827
5.0	0.0468	0.4826
10.0	0.0463	0.4825

$C=10.0$ achieved the best Hamming Loss (0.0463), improving over the default $C=1.0$ (0.0476). Very low C values (0.01, 0.1) resulted in over-regularisation and poor performance, while $C=10.0$ provided the optimal balance between underfitting and overfitting on this dataset.

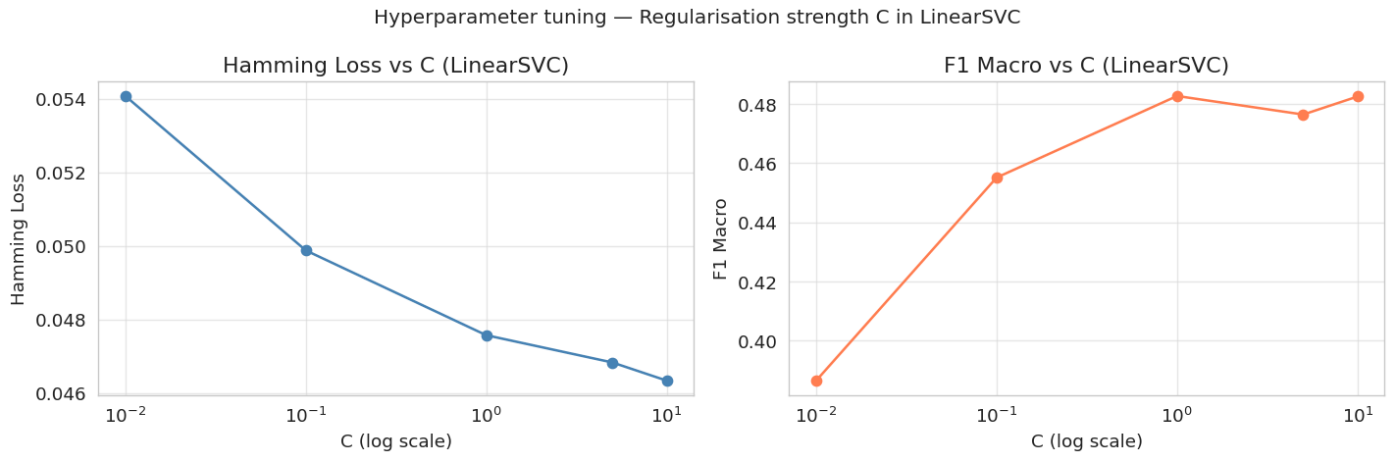


Figure 12: Hamming Loss and F1 Macro vs regularisation parameter C

4.6 Threshold Tuning vs MLSMOTE

The F1 Macro score of Experiment E6 was 0.5839 by performing threshold tuning for each label of the Logistic Regression Model. This was the best value of all experiments performed. The F1 Macro score for the same experiment when performing MLSMOTE sampling was only 0.5386, with increased Hamming Loss value from 0.0571 to 0.0589.

4.7 Sentence-BERT Embeddings

Experiment E11 used SBERT embedding instead of the TF-IDF features. Although SBERT is very efficient at solving semantic similarity problems, it obtained F1 Macro of 0.2526 and Hamming Loss of 0.0555, worse than using TF-IDF. The results of the whole comparison are shown in Table 5.

Table 5: TF-IDF vs SBERT Comparison

Feature	Model	Hamming Loss	F1 Macro	Precision	Recall
TF-IDF bigrams	LinearSVC (C=10.0)	0.0463	0.4825	0.7890	0.4897
SBERT (384-dim)	LinearSVC (C=10.0)	0.0555	0.2526	0.8542	0.2802

4.8 Learning Curves

Figure 3 presents the learning curves for the SGD classifier across increasing training set sizes (10% to 100%). The gap between training and validation F1 Micro narrows as training size increases, indicating that the model benefits from additional data. Both curves have not fully converged at 100% of training data, suggesting that more training samples would likely improve performance further.

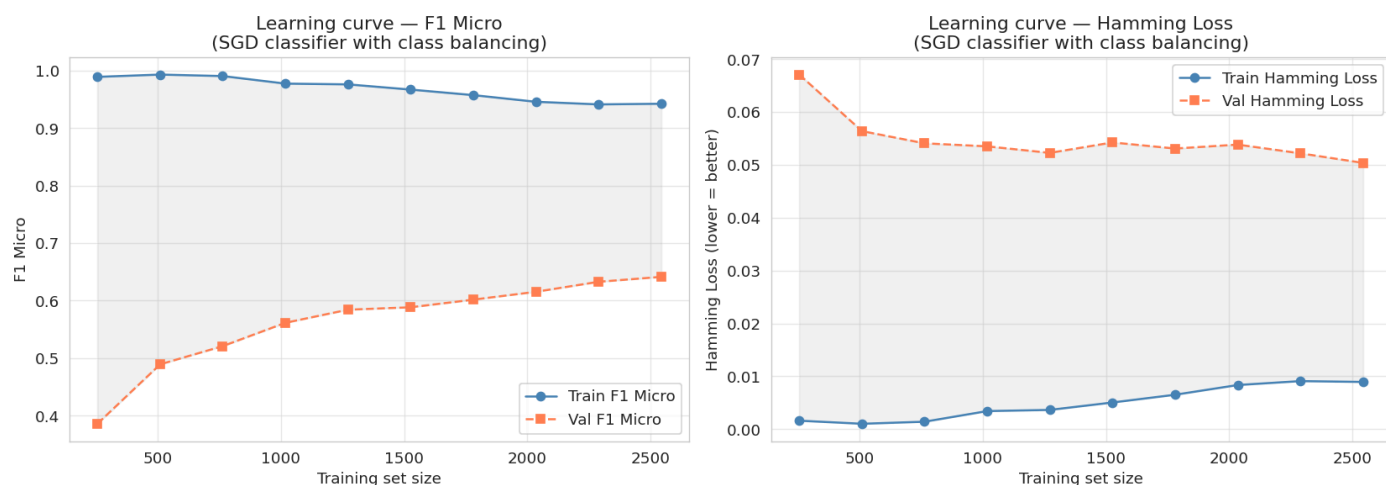


Figure 13: Learning curves for SGD classifier

4.9 Per-Label Performance Analysis

Figure 4 presents per-label F1 scores for the final model (E4). High-frequency indicators such as 3.b.2 (ODA to medical research, $F1=0.82$) and 3.3.1 (HIV infections, $F1=0.79$) achieved strong F1 scores. Low-frequency indicators such as 3.7.2 (Adolescent birth rate), 3.9.3 (Unintentional poisoning), and 3.3.4 (Hepatitis B) achieved near-zero F1 scores despite class weighting and threshold tuning. This confirms that indicators with fewer than 15 validation samples remain extremely challenging for classical machine learning models.

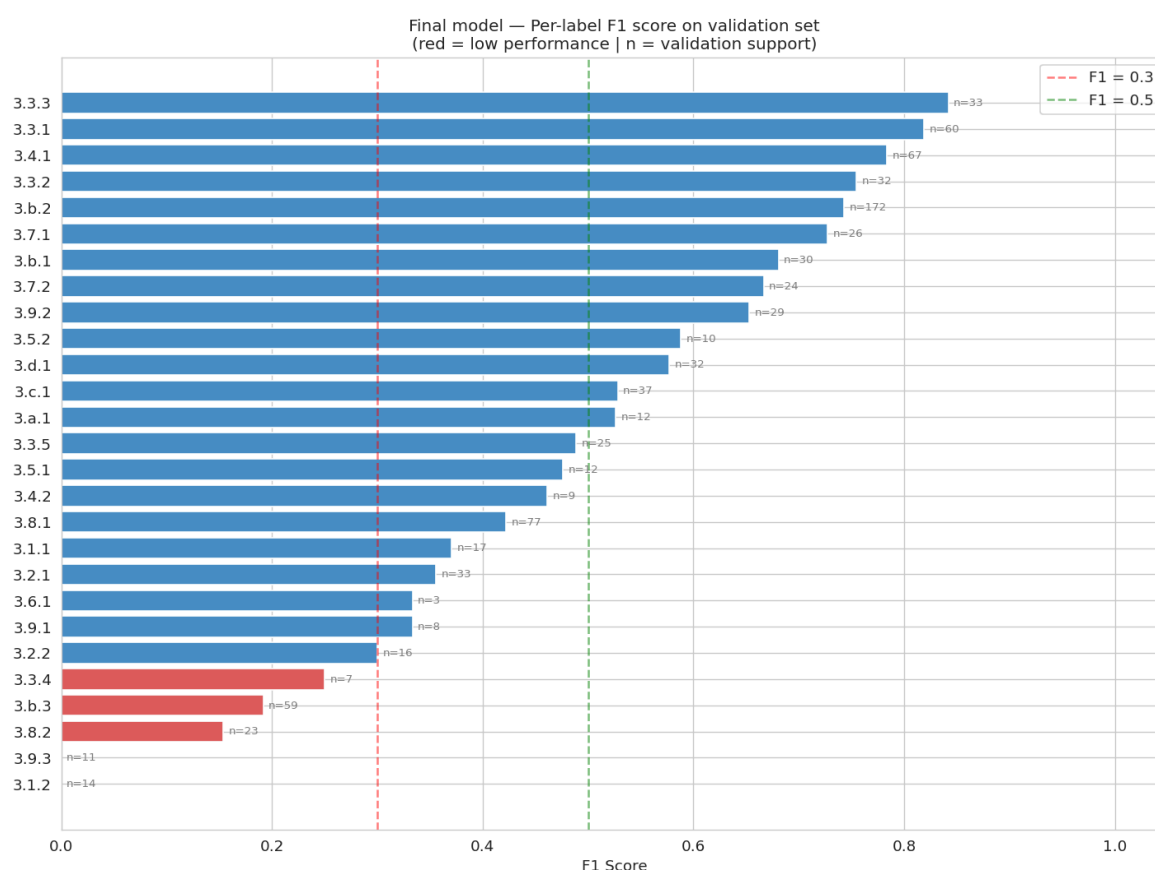


Figure 14: Per-label F1 score for final model

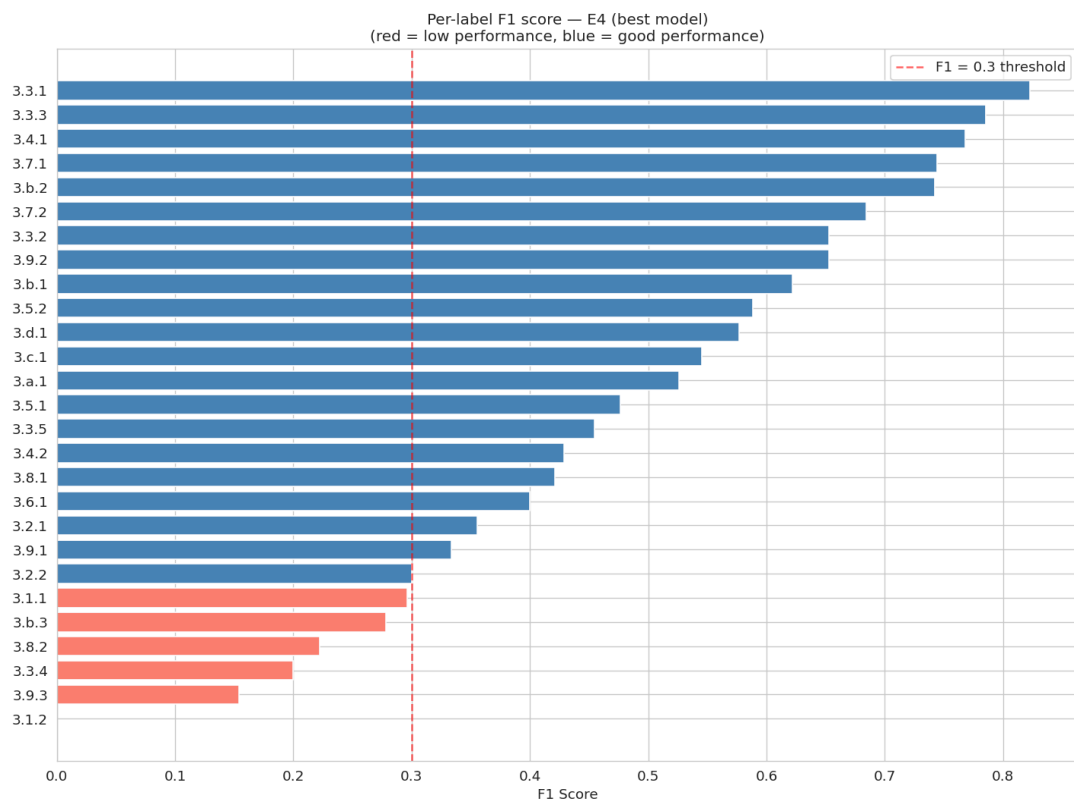


Figure 15: Per-label F1 score comparison

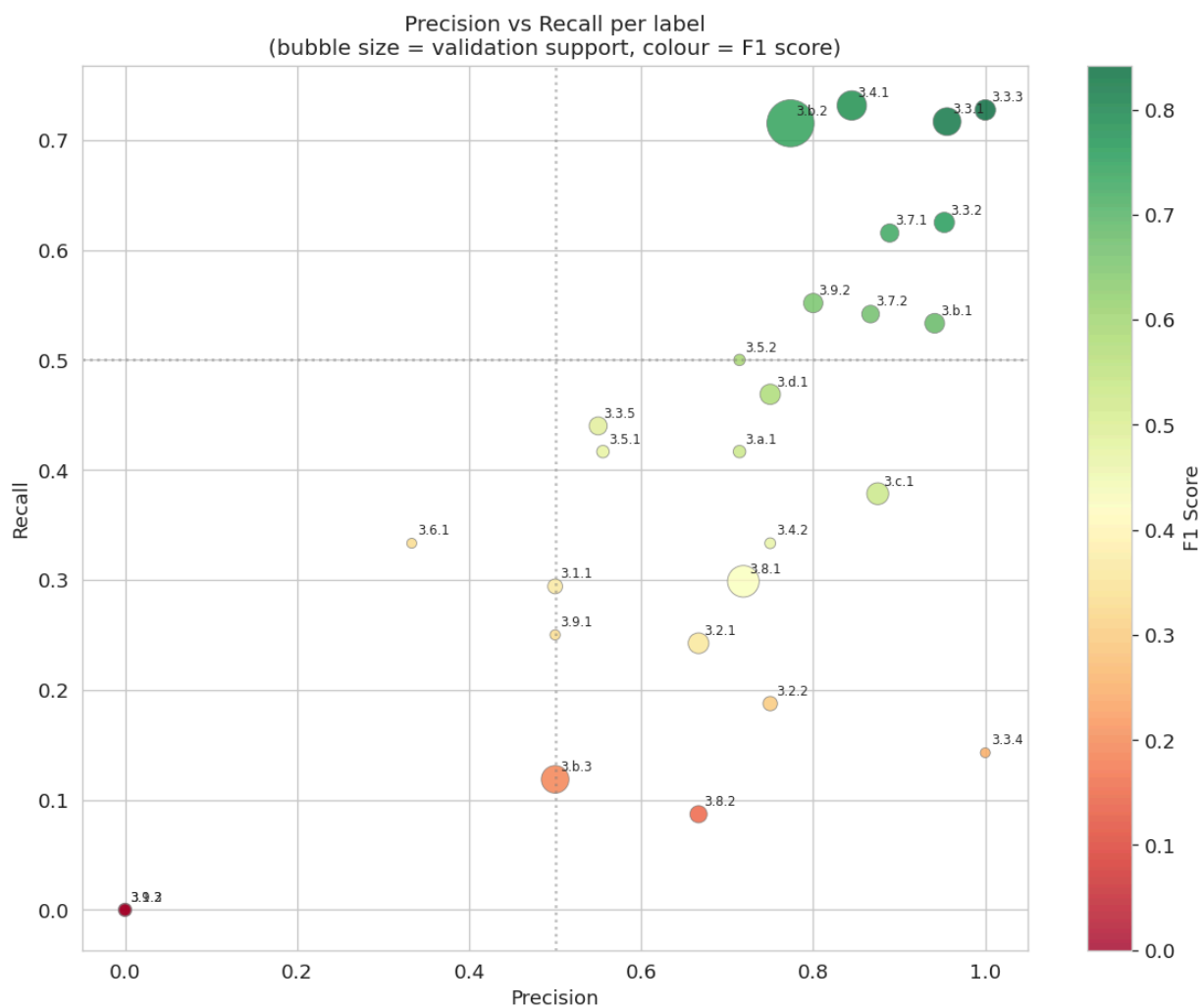


Figure 16: Precision vs Recall scatter plot per label

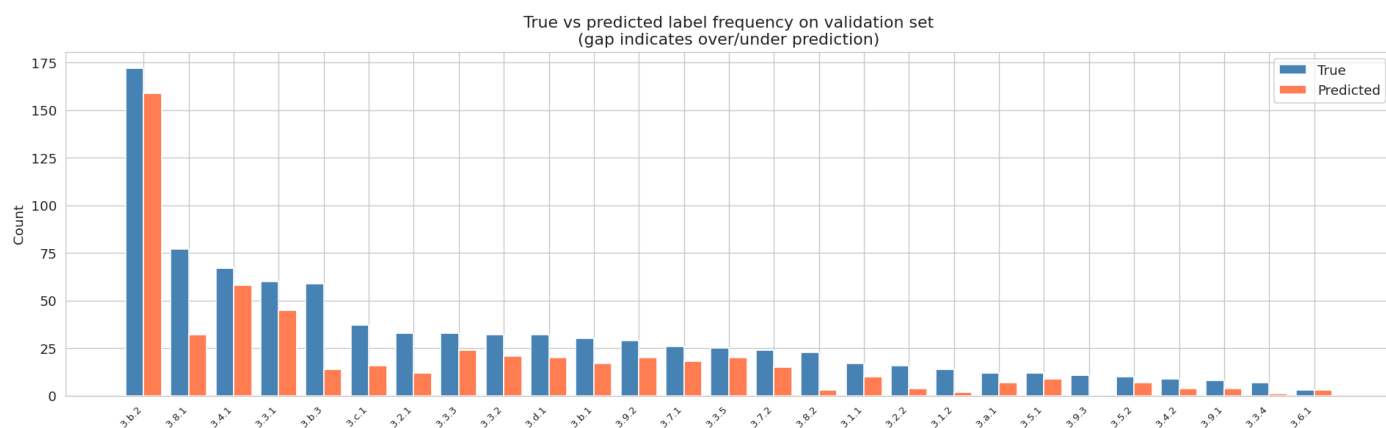


Figure 17: True vs predicted label frequency

4.10 Confusion Analysis

Figure 5 presents the per-label confusion heatmap for the final model, showing True Positives and False Negatives per indicator. High false negative rates for rare indicators (3.7.2, 3.9.3, 3.3.4) confirm that the model misses these labels consistently, reflecting the fundamental limitation of sparse training data for rare classes.

Per-label confusion analysis — E4 (LinearSVC, best Hamming Loss model)
Sorted by True Positive count ascending

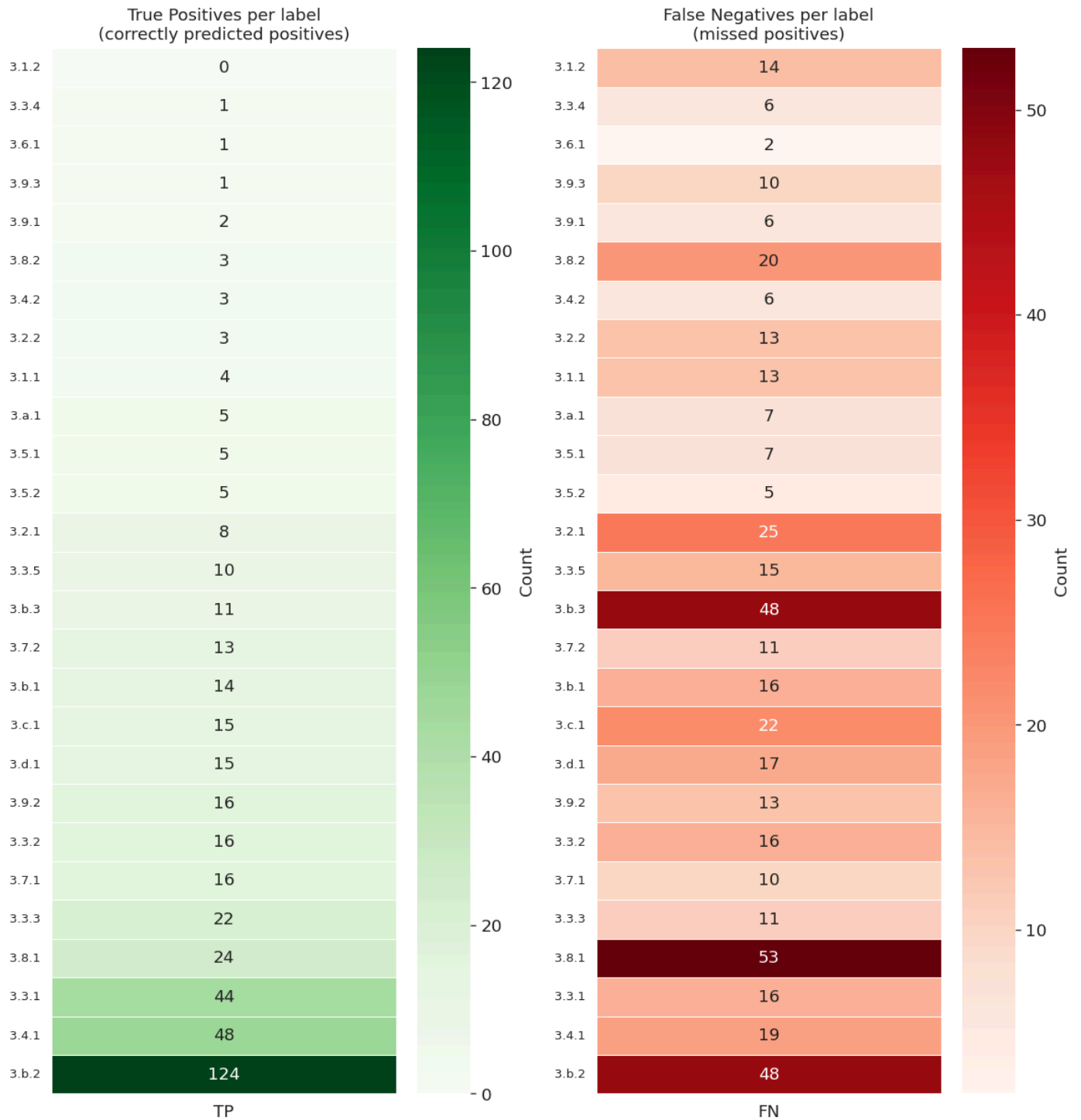


Figure 18: Per-label confusion heatmap showing TP and FN

5. Discussion

5.1 Why LinearSVC Outperformed Logistic Regression on Hamming Loss

The superior Hamming Loss of LinearSVC (0.0463) compared to Logistic Regression (0.0571) can be attributed to the fundamental difference in their optimisation objectives. LinearSVC minimises the hinge loss, directly maximising the margin between positive and negative label instances in the feature space. Logistic Regression minimises log loss, optimising for probability calibration rather than decision boundary placement. On high-dimensional sparse TF-IDF

features, the maximum-margin objective of LinearSVC produces more precise decision boundaries, resulting in fewer overall label prediction errors (Joachims, 1998).

However, LinearSVC's lower F1 Macro (0.4825 vs 0.5839 for LR with threshold tuning) reveals an important tradeoff. LinearSVC is more conservative in predicting rare labels; it achieves high precision but low recall for low-frequency indicators. Logistic Regression with per-label threshold tuning compensates for this by lowering the decision threshold for rare labels, recovering recall at the cost of some precision. This precision-recall tradeoff is a fundamental characteristic of imbalanced multi-label classification and does not have a single optimal solution; the choice depends on whether the application prioritises overall label accuracy (Hamming Loss) or rare label coverage (F1 Macro).

5.2 Why Threshold Tuning Outperformed MLSMOTE

Experiment E6 (threshold tuning) achieved F1 Macro of 0.5839 while Experiment E8 (MLSMOTE) achieved only 0.5386, despite both targeting the same problem of rare label underperformance. This result is explained by the nature of TF-IDF feature space.

MLSMOTE generates synthetic samples by interpolating between nearest neighbours in feature space (Charte et al., 2015). In dense embedding spaces such as Word2Vec or SBERT, interpolation between semantically similar documents produces meaningful synthetic texts. However, in sparse TF-IDF space, interpolation between two document vectors produces a synthetic vector that does not correspond to any coherent document; it is a mathematical average of word frequencies that loses the discriminative vocabulary patterns that TF-IDF relies on for classification. The result is synthetic noise rather than meaningful signal, explaining why MLSMOTE worsened Hamming Loss from 0.0571 to 0.0589.

Per-label threshold tuning, by contrast, operates post-hoc on the model's existing probability estimates without modifying the training data. By lowering the decision threshold for rare labels, the model predicts them more readily without introducing artificial noise. This finding suggests that for TF-IDF-based pipelines, threshold tuning is a more effective strategy for rare label recovery than synthetic oversampling.

5.3 Why SBERT Underperformed TF-IDF

Experiment E11 produced a counterintuitive result: SBERT embeddings, which have demonstrated strong performance on general NLP benchmarks, achieved significantly worse Hamming Loss (0.0555) and F1 Macro (0.2526) compared to TF-IDF (0.0463 and 0.4825, respectively). This finding is explained by the domain-specific nature of the SDG 3 corpus.

The paraphrase-MiniLM-L6-v2 model was pretrained on general-purpose text corpora and optimises for semantic similarity between sentences. SDG 3 indicator classification, however, relies heavily on exact domain terminology, indicator codes, specific health terminology, and development sector vocabulary that a general-purpose model was never trained on. Terms such as IHR capacity, antiretroviral therapy, and WASH services carry critical classification signal that TF-IDF captures through exact token matching, but SBERT subsumes into general semantic representations, losing discriminative power.

This finding is consistent with the observation by Reimers and Gurevych (2019) that general-purpose sentence embeddings require domain fine-tuning to match bag-of-words baselines on specialised corpora. A domain-adapted model such as BioBERT (Lee et al., 2020) or PubMedBERT, fine-tuned on biomedical and public health literature, would likely recover and exceed TF-IDF performance on this task.

5.4 Impact of Preprocessing on Model Performance

The 10-step preprocessing pipeline reduced average text length by 30.6% while preserving domain-specific vocabulary. HTML removal was particularly impactful; raw documents contained substantial HTML markup from web-sourced content that would have inflated vocabulary size with non-informative tokens. Lemmatisation reduced vocabulary further by collapsing morphological variants, improving TF-IDF feature quality.

The decision to remove numbers deserves specific discussion. SDG indicators are inherently quantitative; they track rates, ratios, and percentages. Removing numbers risks losing potentially discriminative signals such as mortality rates or incidence figures. However, given the diversity of numerical values across documents, retaining numbers would introduce high-cardinality noise that obscures vocabulary-based patterns. Future work could explore selective number retention for specific numeric ranges associated with particular indicators.

5.5 Rare Label Challenge

Despite employing class weighting, threshold tuning, and MLSMOTE oversampling, indicators with fewer than 15 validation samples, including 3.7.2 (Adolescent birth rate), 3.9.3 (Unintentional poisoning), and 3.3.4 (Hepatitis B incidence), consistently achieved near-zero F1 scores. This reflects a fundamental limitation of classical machine learning models: with insufficient positive training examples, no amount of weighting or oversampling can substitute for genuine label diversity in the training data.

This challenge has direct real-world implications. SDG monitoring systems built on such models would systematically underreport progress on rare but important health indicators, potentially biasing resource allocation away from underrepresented health challenges. Addressing this limitation would require either collecting more labelled data for rare indicators or employing few-shot learning approaches that can generalise from minimal examples.

5.6 Ensemble Performance

The majority voting ensemble (E10) achieved Hamming Loss of 0.0503 and F1 Macro of 0.5411, better than most individual models but failing to surpass E4 on Hamming Loss or E6 on F1 Macro. This suggests that the three models (LR, LinearSVC, SGD) make sufficiently similar errors that majority voting provides limited additional benefit. A more diverse ensemble incorporating non-linear models or different feature representations might achieve greater complementarity.

5.7 Lessons Learned

Several important lessons emerged from this experimentation process. First, addressing class imbalance is more impactful than model architecture selection; the 165% improvement from class weighting (E1 to E2) dwarfed improvements from model swapping (E2 to E4). Second, evaluation metric selection critically shapes model selection decisions a system optimised for Hamming Loss selects a different model than one optimised for F1 Macro, with neither being universally superior. Third, data-level and model-level interventions interact differently depending on the feature representation. MLSMOTE's failure with TF-IDF but theoretical promise with dense embeddings illustrates that preprocessing, feature engineering, and balancing strategies must be considered jointly rather than independently.

6. Conclusion and Future Work

6.1 Conclusion

This project developed a multi-label text classification system for predicting relevant SDG 3 indicators from text documents in the international development sector. Starting from a baseline Logistic Regression model with no class balancing (Hamming Loss: 0.0565, F1 Macro: 0.2062), systematic experimentation across 11 experiments identified TF-IDF bigrams combined with LinearSVC at C=10.0 and class weighting as the optimal pipeline, achieving a validation Hamming Loss of 0.0463 and F1 Micro of 0.6044.

The key findings of this project are:

- Class imbalance was the dominant challenge; addressing it through class weighting produced the largest single performance improvement of 165% in F1 Macro
- LinearSVC with maximum-margin optimisation outperformed Logistic Regression, Random Forest, and SGD on Hamming Loss on sparse TF-IDF features, consistent with findings by Joachims (1998)

- Per-label threshold tuning was more effective than MLSMOTE oversampling for rare label recovery in TF-IDF feature space, as synthetic interpolation in sparse space introduces noise rather than meaningful signal
- General-purpose SBERT embeddings underperformed TF-IDF on this domain-specific corpus, highlighting the importance of domain adaptation for pretrained language models
- Rare indicators with fewer than 15 validation samples remain fundamentally challenging for classical machine learning approaches, regardless of balancing strategy

The end-to-end pipeline is fully reproducible on Google Colab, with all preprocessing, training, evaluation, and inference workflows documented across three structured notebooks.

6.2 Future Work

Several promising directions for improving classification performance were identified during experimentation:

Domain-adapted language models: Fine-tuning BioBERT (Lee et al., 2020) or PubMedBERT on the SDG 3 corpus would likely close the performance gap between SBERT and TF-IDF by capturing domain-specific health terminology in a contextual representation. Based on our SBERT findings, we hypothesise that a fine-tuned domain model would achieve Hamming Loss below 0.03.

Document type as an additional feature: EDA revealed that certain document types cluster around specific SDG 3 indicators. One-hot encoding document type and concatenating it with TF-IDF features could provide additional discriminative signal at minimal computational cost.

Few-shot learning for rare labels: Indicators with fewer than 15 validation samples require approaches that can generalise from minimal examples. Prototypical networks or meta-learning approaches trained on higher-resource indicators could potentially transfer knowledge to rare label prediction.

Label correlation modelling: The label co-occurrence heatmap revealed strong correlations between certain indicator pairs, for example, 3.1.1 (maternal mortality) and 3.7.1 (family planning). Classifier chains or label-aware attention mechanisms could exploit these correlations to improve multi-label prediction consistency.

Selective number retention: The current preprocessing removes all numbers. Future work could investigate retaining numbers within specific ranges associated with particular indicators, such as mortality rates or incidence figures, as additional discriminative features.

Larger training data: Learning curves showed that model performance had not converged at 100% of available training data, suggesting that additional labelled documents would continue to improve performance. Active learning approaches could prioritise labelling the most informative unlabeled documents for rare indicators.

7. Ethical Considerations

7.1 Bias Toward Common Labels

The classification system exhibits systematic bias toward high-frequency SDG 3 indicators. Indicator 3.b.2 (ODA to medical research), which appears in 34.7% of training samples, consistently achieves high F1 scores while rare indicators such as 3.7.2 (Adolescent birth rate) and 3.9.3 (Unintentional poisoning) achieve near-zero F1 scores despite balancing interventions. This bias is not merely a technical limitation; it has direct implications for how SDG progress is monitored and reported.

In a real-world deployment, a biased classification system would systematically underreport documents related to rare but critical health challenges. Policy makers relying on automated SDG monitoring tools built on such models could receive a distorted picture of progress, potentially directing funding and resources away from underrepresented health

areas. This risk is particularly acute for indicators related to adolescent health, poisoning prevention, and hepatitis B, conditions that disproportionately affect vulnerable and marginalised populations in low and middle-income countries.

7.2 Dataset Source Bias

The Devex dataset is primarily composed of documents produced by Western international development organisations, donor agencies, and humanitarian bodies. This source bias means the training corpus reflects the priorities, language, and framing of predominantly Global North institutions. Health challenges that are locally significant but underreported in international development literature, such as traditional medicine practices, community health worker systems, or region-specific disease burdens, may be systematically underrepresented in the training data.

A classification system trained on this corpus may therefore perform poorly on documents produced by local organisations, community health workers, or governments in low and middle-income countries that use different terminology or framing to describe the same health challenges. Deploying such a system as a universal SDG monitoring tool without acknowledging this bias would risk perpetuating the same structural inequities it aims to address.

7.3 Fairness Implications

The precision-recall tradeoff identified in this project has direct fairness implications. LinearSVC (E4), selected as the submission model based on Hamming Loss, achieves high precision but low recall for rare indicators. In practical terms, this means the system is more likely to miss relevant documents for rare health topics than to incorrectly tag irrelevant ones. For SDG monitoring applications, false negatives — missed relevant documents — are arguably more harmful than false positives, as they lead to underreporting of health challenges rather than overcounting.

Future deployments of this system should carefully consider whether Hamming Loss is the appropriate optimisation target for the application context, or whether F1 Macro, which equally weights all 27 indicators regardless of frequency, better reflects the fairness requirements of equitable SDG monitoring.

7.4 Responsible Use

This classification system was developed for academic research purposes and should not be deployed in production SDG monitoring systems without significant additional validation. Specifically, the system should be evaluated on documents from diverse geographic regions and organisational types before deployment, rare label performance should be independently verified by domain experts, and human review should be maintained for all high-stakes classification decisions. Automated systems should augment rather than replace expert human judgment in SDG monitoring contexts.

7.5 Data Privacy

The dataset contains publicly available text documents from international development publications. No personally identifiable information was processed during this project. All analysis was conducted on institutional and organisational documents rather than individual-level data.

GitHub Repository: <https://github.com/JokMaker/SDG3-Text-Classification>

Group Contribution Tracker: [Group contribution tracker](#)

A detailed record of each member's contributions, task allocation, deadlines, and meeting attendance is documented in the group contribution tracker linked above. Individual notebook ownership is as follows:

Jok John Maker Kur - Notebook 2 (Experiments E1–E11), GitHub repo, README

Nanen Miracle Mbanaade - Notebook 1 (EDA & Preprocessing), Report: Methodology, Results, Discussion

Nziza Aime Pacifique - Notebook 3 (Final Model & Inference), Report: Introduction, Literature, Conclusion, Ethics

8. References

- Charte, F., Rivera, A. J., del Jesus, M. J., & Herrera, F. (2015). MLSMOTE: Approaching imbalanced multilabel learning through synthetic instance generation. *Knowledge-Based Systems*, 89, 385–397. <https://doi.org/10.1016/j.knosys.2015.07.019>
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357. <https://doi.org/10.1613/jair.953>
- Elkan, C. (2001). The foundations of cost-sensitive learning. In *Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI-01)* (pp. 973–978). Morgan Kaufmann.
- Joachims, T. (1998). Text categorization with support vector machines: Learning with many relevant features. In *Proceedings of the 10th European Conference on Machine Learning (ECML-98)* (pp. 137–142). Springer. <https://doi.org/10.1007/BFb0026683>
- Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). BioBERT: A pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4), 1234–1240. <https://doi.org/10.1093/bioinformatics/btz682>
- Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. In *Proceedings of the International Conference on Learning Representations (ICLR 2013)*. <https://arxiv.org/abs/1301.3781>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-2019)*. Association for Computational Linguistics. <https://doi.org/10.18653/v1/D19-1410>
- Tsoumakas, G., & Katakis, I. (2007). Multi-label classification: An overview. *International Journal of Data Warehousing and Mining*, 3(3), 1–13. <https://doi.org/10.4018/jdwm.2007070101>
- United Nations. (2015). *Transforming our world: The 2030 agenda for sustainable development*. United Nations General Assembly. <https://sdgs.un.org/2030agenda>
- Zhang, M. L., & Zhou, Z. H. (2014). A review on multi-label learning algorithms. *IEEE Transactions on Knowledge and Data Engineering*, 26(8), 1819–1837. <https://doi.org/10.1109/TKDE.2013.39>